CI/CD Pipelines: Powering Machine Learning Projects

By embracing continuous integration and continuous deployment, machine learning systems can keep pace with the environments they operate in. GitHub Actions helps build automation into the workflow, leading not only to a well-performing model but a complete pipeline that continuously delivers value to users.

Machine learning is now a core part of everyday applications. Whether it is personalising what we watch, detecting banking frauds, or improving health diagnostics, ML systems are constantly learning and improving. But behind every successful model that reaches users lies a demanding process. Training a model in a notebook is only the beginning. To work in the real world, the model must be tested carefully and deployed in a way that is repeatable, reliable, and fast.

This is where CI/CD, short for continuous integration and continuous deployment, becomes important. These practices transformed the software development world by automating testing and deployment every time the code changed. For machine learning, they do even more. Along with code, ML projects must manage data, experiments, and trained model files, all of which change over time. A model that performs wonderfully one week may suddenly fail if the data it sees begins to shift. Without automation, it becomes difficult to trust that each new update is actually improving the model.

GitHub Actions has made CI/CD more accessible than ever for ML teams. Because it is built into GitHub, automation can attach directly to the same place where code and models are versioned. With a simple workflow file, tasks like installing dependencies, verifying data pipelines, checking metrics, running lightweight training, and deploying updated models can all happen automatically. This removes the repeated manual steps that often lead to mistakes or inconsistent environments.

By using CI/CD, machine learning development becomes more confident and organised. Each change is tested before it reaches production. Every model has a clear history of how it was built and which data it used. Deployments become routine rather than risky. Most importantly, teams can deliver improvements faster, ensuring the model stays accurate as the world around it changes. Instead of a one-off experiment, machine learning grows into a continuous, sustainable solution.

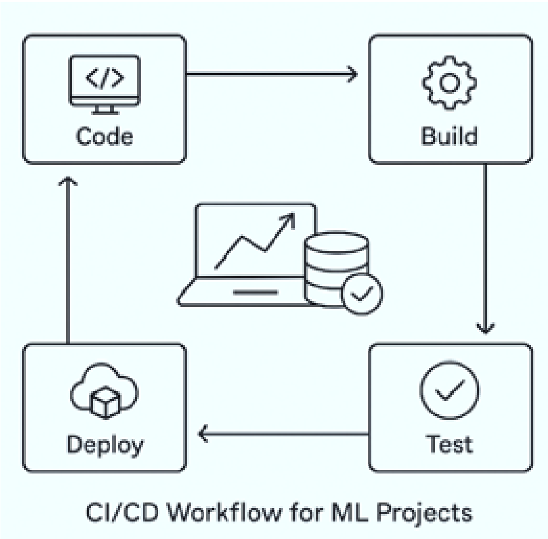

Figure 1: Introduction to CI/CD for machine learning

Understanding the ML project lifecycle

Every machine learning project follows a journey long before a model ever reaches users. It begins with gathering data, often from multiple messy sources that require cleaning and organisation. That data then goes through careful preparation, where missing values are handled and meaningful features are created. Only after this groundwork is done can model training truly begin.

Training is rarely a one-time process. Developers try different algorithms, tune parameters, compare results, and repeat the cycle until performance meets expectations. When the model seems ready, the work shifts from experimentation to production. The model needs to be packaged along with the code that makes predictions and deployed somewhere it can serve real users, whether through a web API, a cloud server, or even an application running on edge devices.

But deployment is not the finish line. It is more like the starting point of a real test. Once the model interacts with live data, its performance can change. Customer behaviour may shift, noise in sensors might increase, or market conditions can suddenly evolve. If no one keeps an eye on how the model is performing, its accuracy may silently drop and affect users in unexpected